

AI-Powered Darshan ease

Project Description:

Project Description – DarshanEase: Temple Appointment Booking System

DarshanEase is a web-based **Temple Appointment Booking System** developed using the **MERN Stack (MongoDB, Express.js, React.js, and Node.js)**. The application is designed to simplify the process of booking temple darshan appointments by providing an online platform where devotees can register, log in, view temple information, check available darshan slots, and book appointments conveniently from anywhere.

The system eliminates the need for long queues and manual booking procedures by offering a secure and user-friendly digital solution. It supports **role-based access control**, allowing users, organizers, and administrators to perform their respective tasks efficiently. Users can manage their profiles, view booking history, cancel bookings when required, and make online donations to temples. Organizers can manage temple information, available darshan slots, and bookings, while administrators have complete control over user management, temple management, and overall system monitoring.

The backend is built with **Node.js** and **Express.js**, while **MongoDB** is used as the database to securely store user, temple, booking, donation, and slot information. **JWT (JSON Web Token)** is implemented for secure authentication and authorization, ensuring that only authenticated users can access protected resources.

The primary objective of **DarshanEase** is to provide a reliable, efficient, and scalable platform that enhances the temple visiting experience by reducing waiting time, improving appointment management, and enabling seamless online services. The application also demonstrates modern web development practices, including RESTful APIs, secure authentication, responsive user interfaces, and efficient database management, making it a comprehensive solution for digital temple appointment management.

Scenarios

Scenario 1: User Registration

The user opens the DarshanEase application and navigates to the registration page. They enter their name, email address, phone number, and password. After submitting the form, the system validates the entered details, checks whether the email is already registered, encrypts the password, and stores the user's information securely in the database. Upon successful registration, the user receives a confirmation message and can proceed to log in.

Scenario 2: User Login

A registered user enters their email address and password on the login page. The system verifies the credentials against the database. If the credentials are valid, the user is authenticated using JWT and redirected to the home page. If the credentials are invalid, an appropriate error message is displayed.

Scenario 3: View Temple Information

After logging in, the user accesses the list of available temples. The system retrieves temple details from the database and displays information such as the temple name, location, timings, and description. The user can browse and select a temple to view more details.

Scenario 4: View Darshan Slots

The user selects a temple to view the available darshan slots. The system retrieves all active slots associated with the selected temple and displays their date, time, and availability. The user chooses a suitable slot for booking.

Scenario 5: Book Darshan Appointment

The user selects an available darshan slot and submits the booking request. The system verifies slot availability, creates a booking record, updates the available capacity if required, and displays a booking confirmation. The booking details are stored in the database for future reference.

Scenario 6: View Booking History

The user navigates to the "My Bookings" section to view all previous and upcoming darshan appointments. The system retrieves the booking history from the database and displays the booking details, including temple name, booking date, slot time, and booking status.

Scenario 7: Cancel Booking

If the user is unable to attend the scheduled darshan, they can cancel the booking before the specified time. The system updates the booking status to "Cancelled" and makes the slot available for other users if applicable. A confirmation message is displayed after successful cancellation.

Scenario 8: Make a Donation

The user visits the donation page, selects a temple, enters the donation amount, and submits the donation request. The system records the donation details and displays a confirmation message indicating that the donation has been successfully processed.

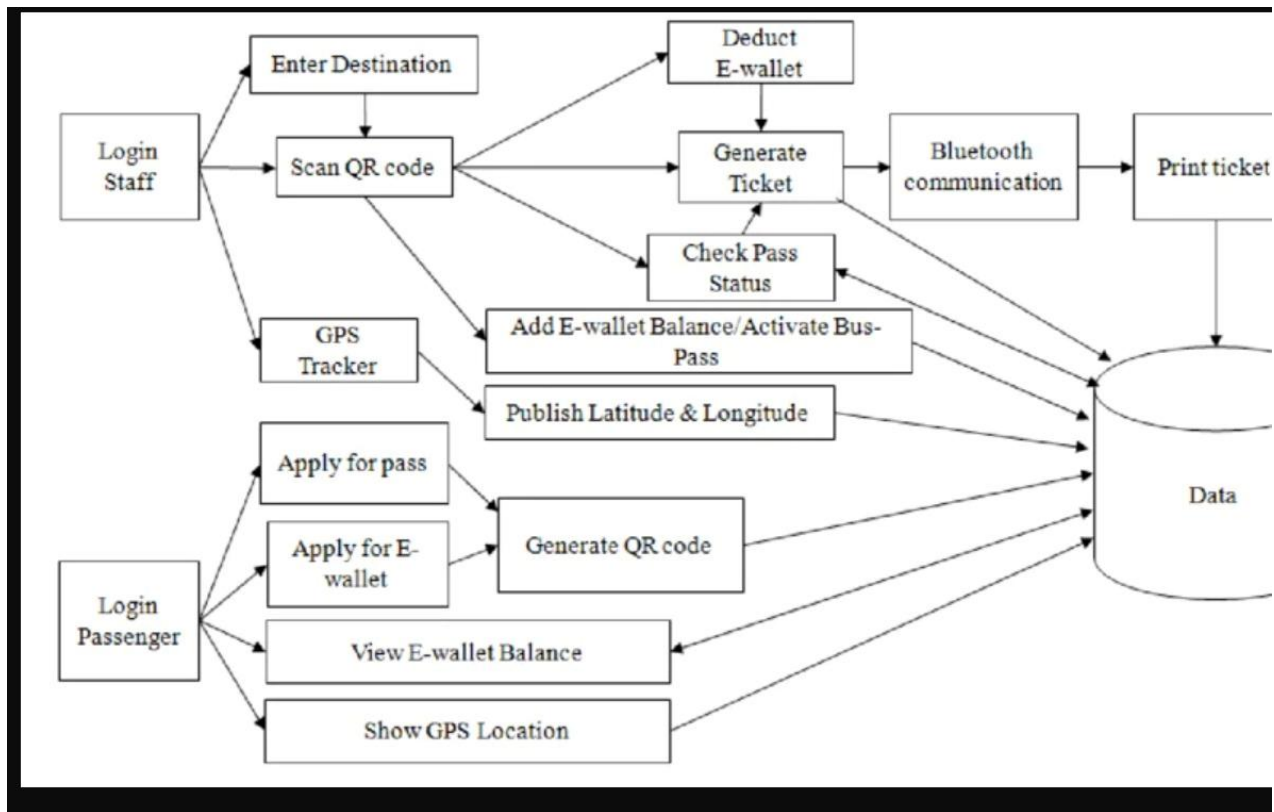
Scenario 9: Manage User Profile

The user opens the profile section to view or update personal information such as name, phone number, or password. The system validates the updated information, saves the changes in the database, and confirms that the profile has been updated successfully.

Scenario 10: Organizer and Admin Management

Organizers log in to manage temple details, darshan slots, and bookings assigned to their temples. Administrators access the admin dashboard to monitor users, bookings, donations, temples, and system activities. The system ensures that only authorized users can access these administrative features through role-based authentication and authorization.

Technical Architecture:



Description: Technical Architecture Description

The DarshanEase – Temple Appointment Booking System is developed using the MERN Stack, which includes MongoDB, Express.js, React.js, and Node.js. The frontend is built with React.js to provide a responsive and user-friendly interface where users can register, log in, browse temple details, book darshan slots, make donations, and manage their profiles. The backend is developed using Node.js and Express.js, which process user requests, implement business logic, and provide secure RESTful APIs. JWT (JSON Web Token) is used for secure user authentication and role-based access control for users, organizers, and administrators.

The application uses MongoDB as the database to store user information, temple details, darshan slots, booking records, and donation details. Mongoose is used to connect the backend with MongoDB and perform database operations efficiently. The frontend communicates with the backend through REST APIs, while the backend retrieves and stores data in the database. The project is developed using Visual Studio Code, tested using Postman and Thunder Client, and managed with Git and GitHub. This architecture provides a secure, scalable, and efficient solution for managing online temple appointments and related services.

(Note: If you are using database in your project definitely you need to add ER Diagram along with description)

Pre-requisites:

- **Node.js Knowledge:** Understanding of Node.js for developing the backend server and handling RESTful APIs.
- **Express.js Framework:** Familiarity with Express.js for API routing, middleware, and server-side application development.
- **React.js Knowledge:** Basic understanding of React.js for building responsive and interactive user interfaces.
- **MongoDB & Mongoose:** Knowledge of MongoDB for database management and Mongoose for schema design and CRUD operations.
- **HTML, CSS, and JavaScript Skills:** Proficiency in frontend technologies for developing responsive web pages and user interfaces.
- **JWT Authentication:** Understanding of JSON Web Token (JWT) for secure user authentication and role-based authorization.
- **Git & GitHub:** Familiarity with Git for version control and GitHub for source code management and collaboration.
- **Visual Studio Code:** Basic knowledge of Visual Studio Code as the primary development environment.
- **API Testing Tools:** Experience with **Postman** or **Thunder Client** for testing RESTful APIs.
- **MongoDB Compass:** Knowledge of MongoDB Compass for viewing, managing, and monitoring the MongoDB database.

These prerequisites are suitable for the **DarshanEase** project and can be included in your TS/project documentation.

Project Workflow:

Milestone 1: Requirement analysis

The project started by identifying the problem statement, gathering user requirements, defining the project objectives, and preparing the Software Requirements Specification (SRS). Functional and non-functional requirements were analyzed to determine the overall scope of the system.

Milestone 2: System Design

The system architecture, database schema, flowcharts, and user interface layouts were designed. The technical design included defining the relationships between users, temples, darshan slots, bookings, and donations.

Milestone 3: Backend Development

The backend was developed using **Node.js**, **Express.js**, and **MongoDB**. RESTful APIs were created for user authentication, temple management, darshan slot management, booking management, donation management, profile management, and administrator functionalities. JWT authentication and role-based authorization were also implemented

Milestone 4: Frontend Development

The frontend was developed using **React.js**. Responsive web pages were created for user registration, login, temple listing, darshan slot booking, donations, profile management, and administrator dashboards, providing a user-friendly experience.

Milestone 5: Deployment

. Finally, the project documentation, testing reports, user manual, and presentation were prepared.

The application was verified for deployment readiness, ensuring that the **DarshanEase – Temple Appointment Booking System** met the project requirements and was ready for demonstration and future deployment.

Milestone 6: Conclusion

The **DarshanEase – Temple Appointment Booking System** is a simple and efficient web application developed using the **MERN Stack**. It allows users to register, log in, view temple details, book darshan slots, make donations, and manage their profiles through an easy-to-use interface. The system helps reduce manual work, saves time, and provides a convenient online booking experience.